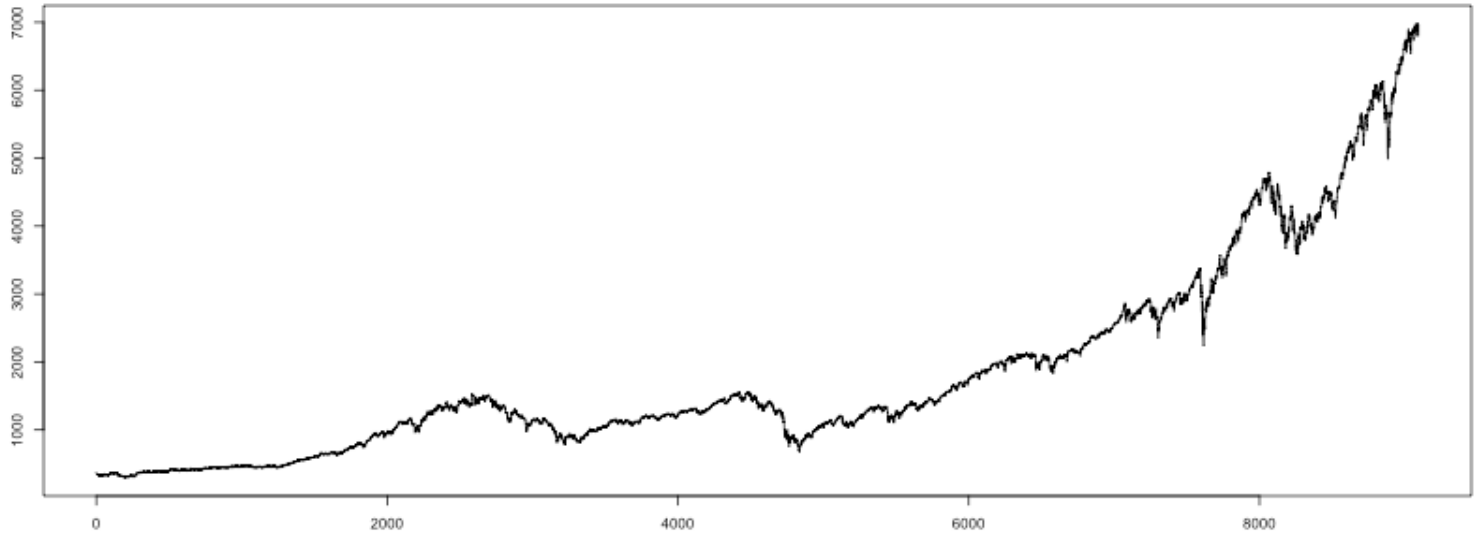
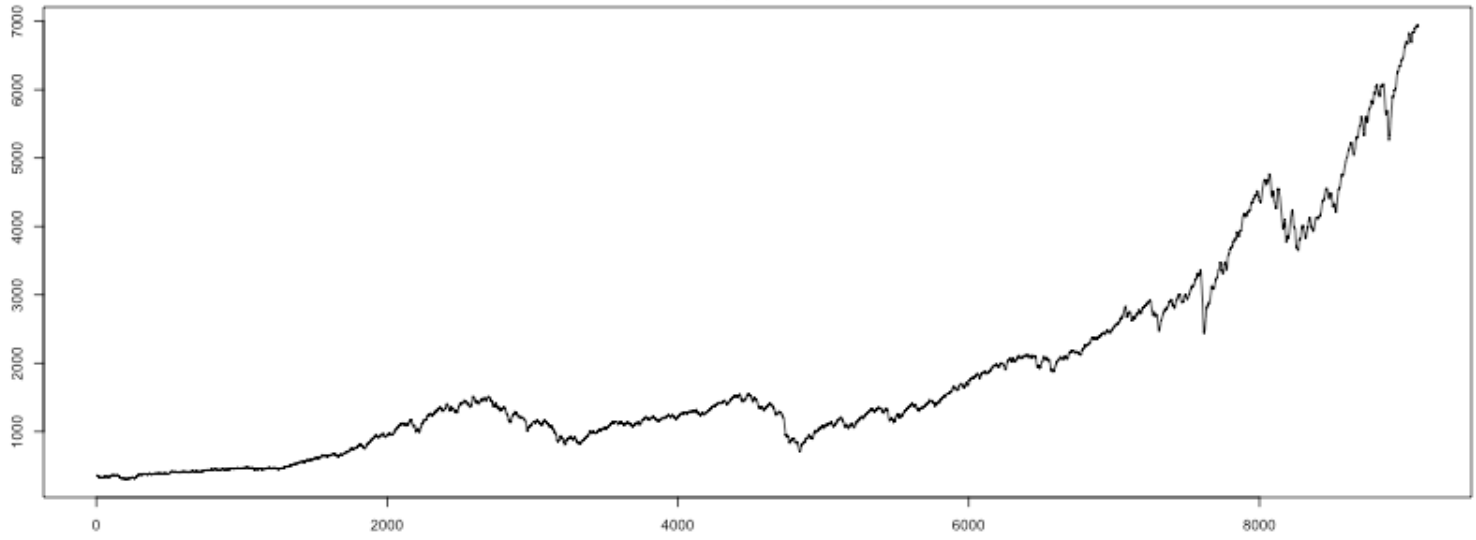
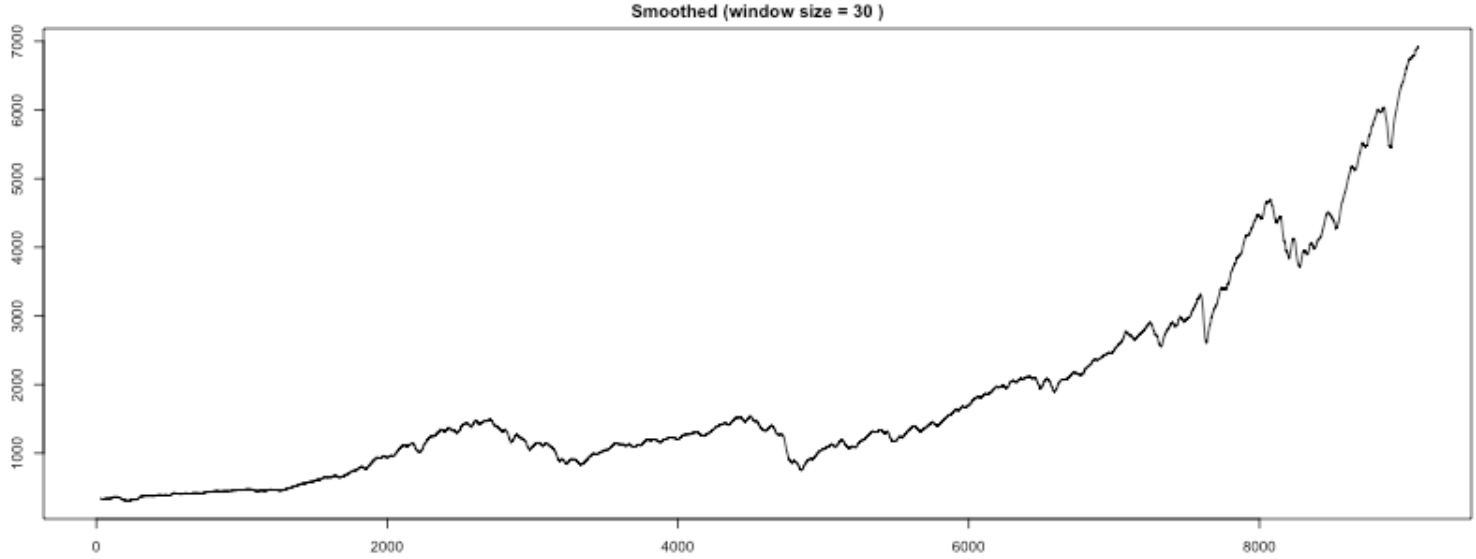
S&P 500 Original Data



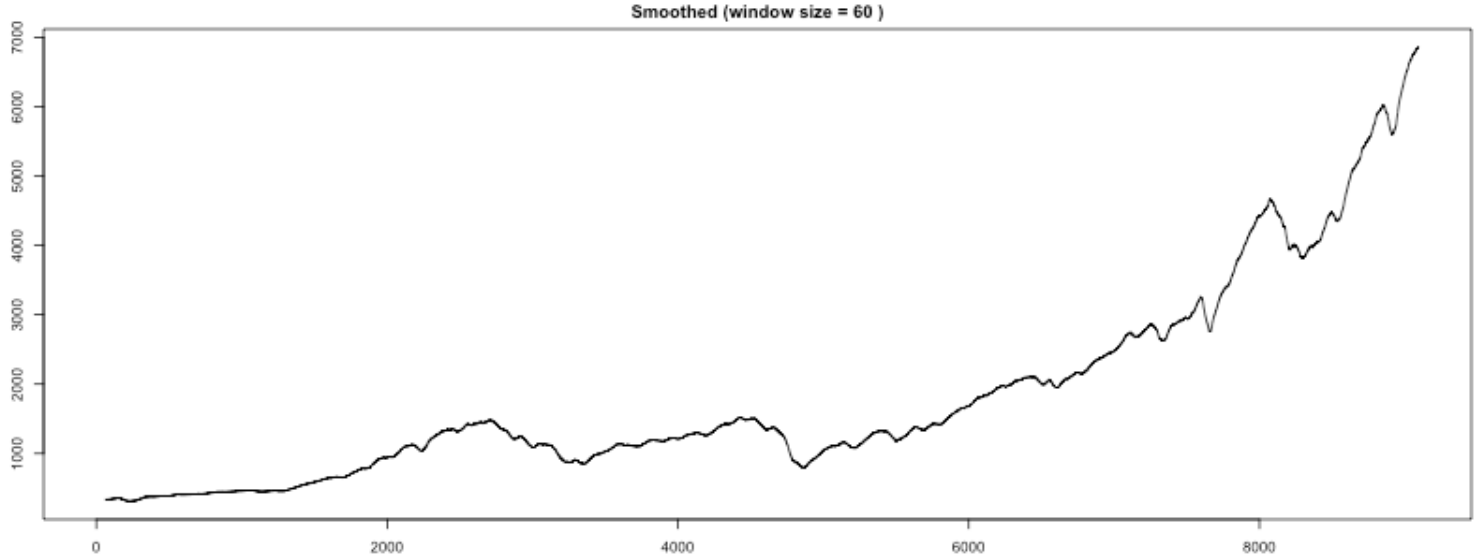
Smoothed (window size = 10)

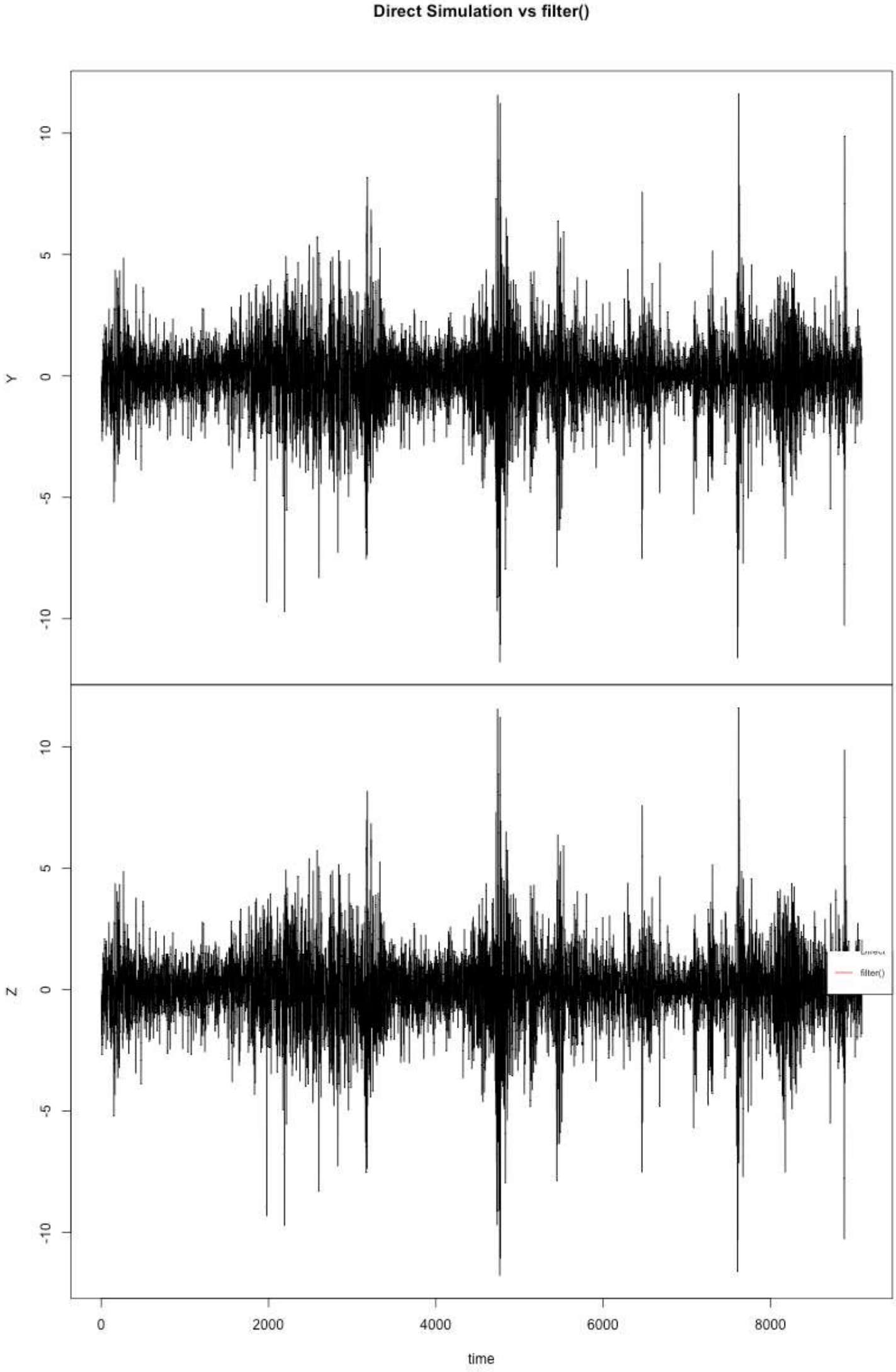


Smoothed (window size = 30)



Smoothed (window size = 60)




Direct Simulation vs filter()
Y
Z
10
5
0
-5
-10
0
2000
4000
6000
8000
time
filter()

hw2_q9.R

Assignments > hw2_q9.R

```r
# Task 9: State Space Model (Problem 3)
# Using S&P 500 Data
library(quantmod)

# Pull S&P 500 data
getSymbols("^GSPC", src = "yahoo", from = "1990-01-01", to = Sys.Date(),
           auto.assign = TRUE, warnings = FALSE)

# Extract closing price and compute returns
X_price = as.numeric(Cl(GSPC))
X = diff(log(X_price)) * 100
X = c(0, X)
N = length(X)

# Problem 3 coefficients (from difference equation)
a1 = 0.9
a2 = -0.5

# State space model
# F = State transition matrix
F = array(c(a1, a2, 1, 0), dim=c(2,2))
# G = Input matrix
G = array(c(1, 0), dim=c(2,1))

cat("State Transition Matrix F:\n")
print(F)
cat("Input Matrix G:\n")
print(G)

# eigenvalues
ev = eigen(F)
cat("\nEigenvalues of F:\n")
print(ev$values)
cat("Absolute values:\n")
print(abs(ev$values))

# Initialize state
Z = array(c(0, 0), dim=c(2,1))
ZM = array(0, dim=c(2,N))

ZM[,1] = F%*%Z + G*X[1]

for (time in 2:N) {
  LZ = ZM[,time-1]
  LZ = array(LZ, dim=c(2,1))
  Z = F%*%LZ + G*X[time]
  ZM[,time] = Z
}

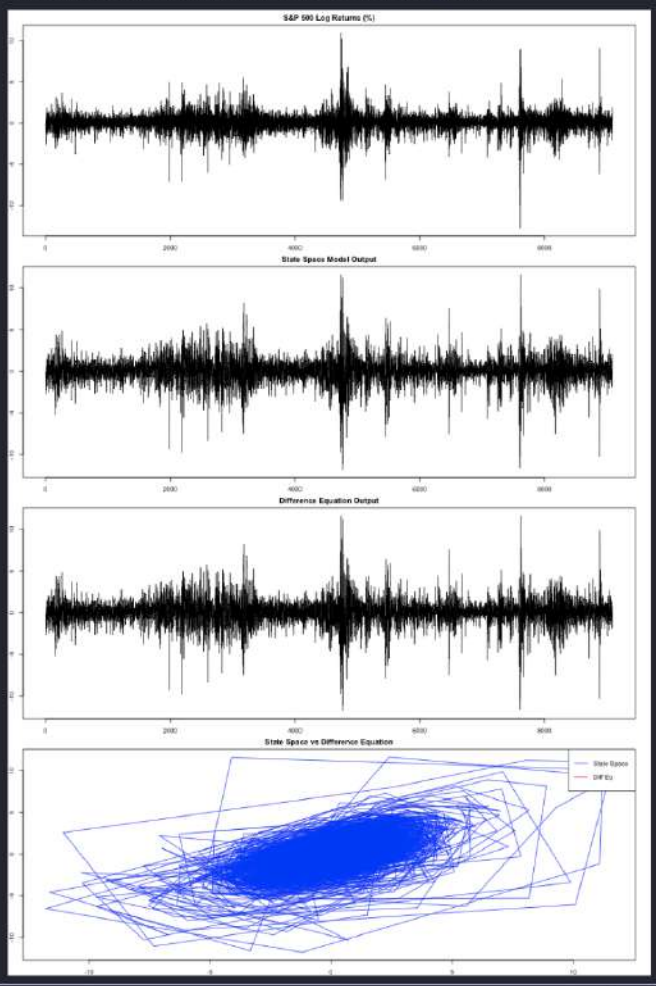
ZM = t(ZM)
Ys = ZM[,1]

# Compare with DE
y0 = 0
y_1 = 0
Y = array(0, dim=c(N,1))

for (index in 1:N) {
  if (index == 1) {yt = a1*y0 + a2*y_1 + 0}
  if (index == 2) {yt = a1*Y[index-1] + a2*y0 + X[index-1]}
  if (index > 2) {yt = a1*Y[index-1] + a2*Y[index-2] + X[index-1]}
  Y[index] = yt
```

plot.png

var > folders > 0v > rd7dv3r524n7kh8f8fqcc0sc0000gn > T > RtmpcEZ02P > vscode-R > plot.png

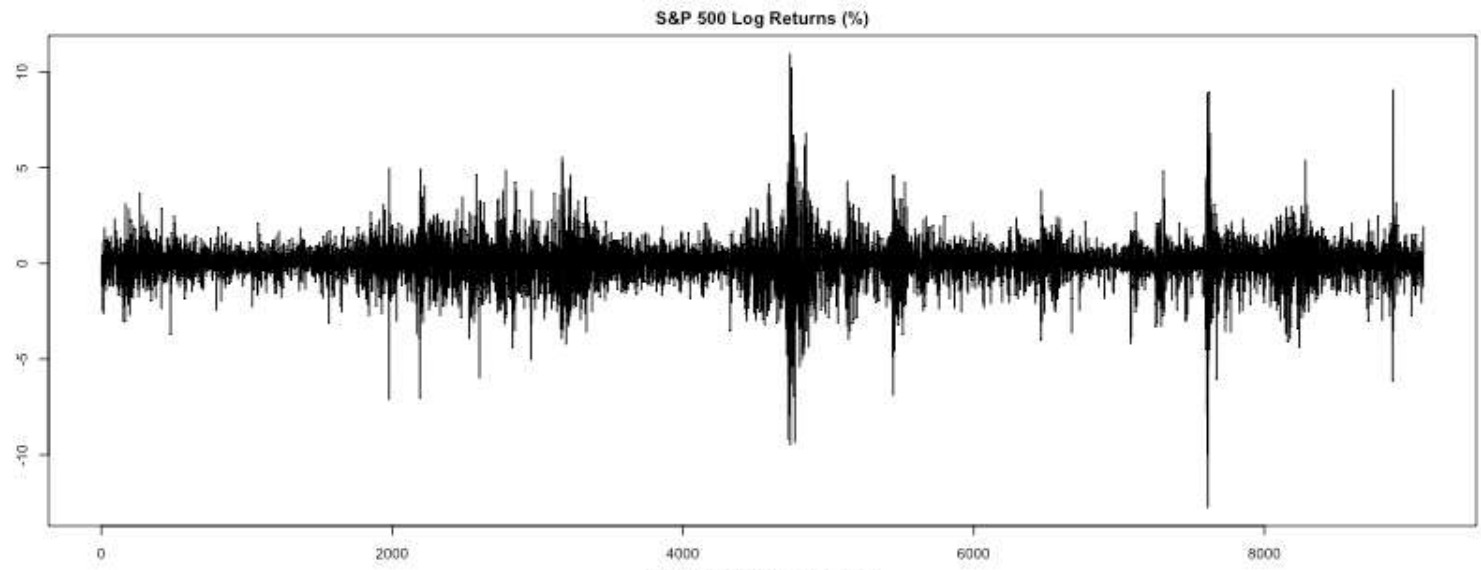


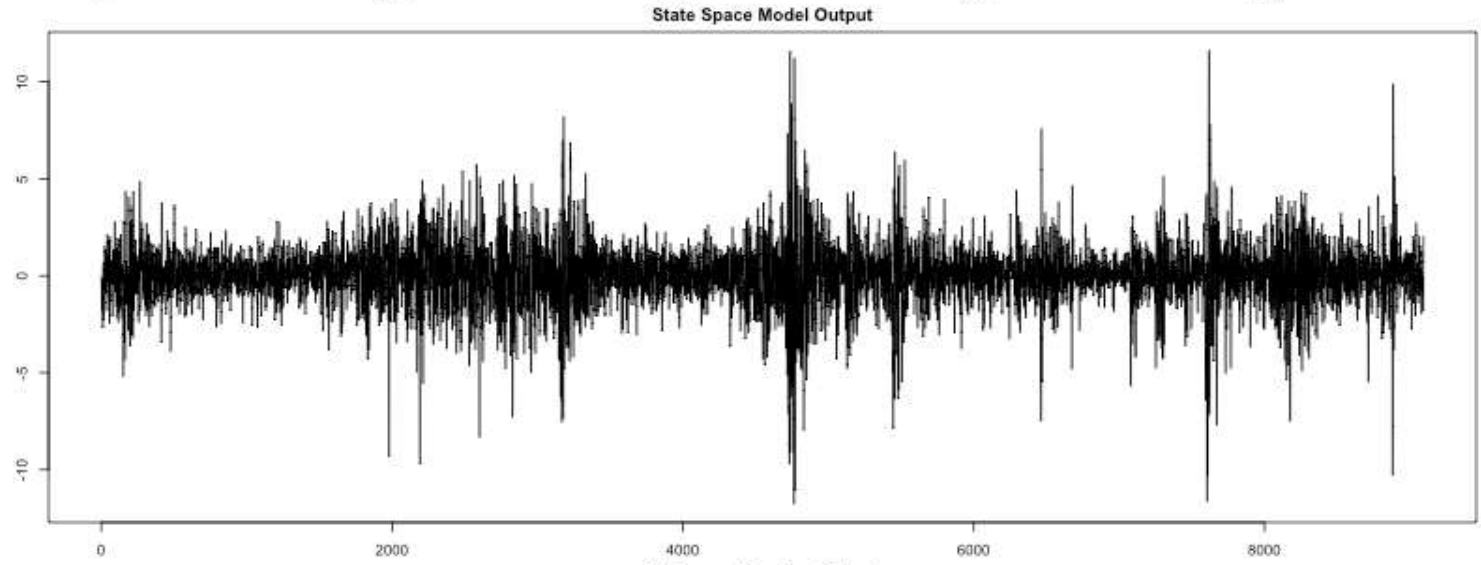
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

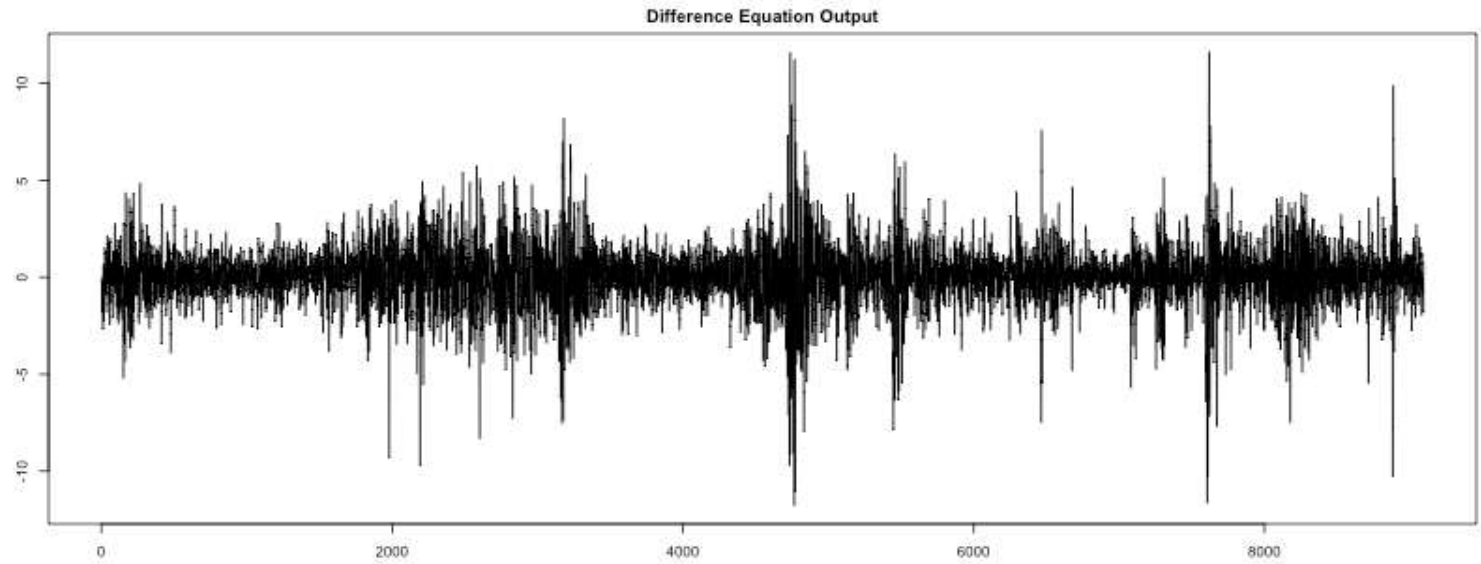
```
Eigenvalues of F:
[1] 0.45+0.54543561 0.45-0.54543561
Absolute values:
[1] 0.7071068 0.7071068

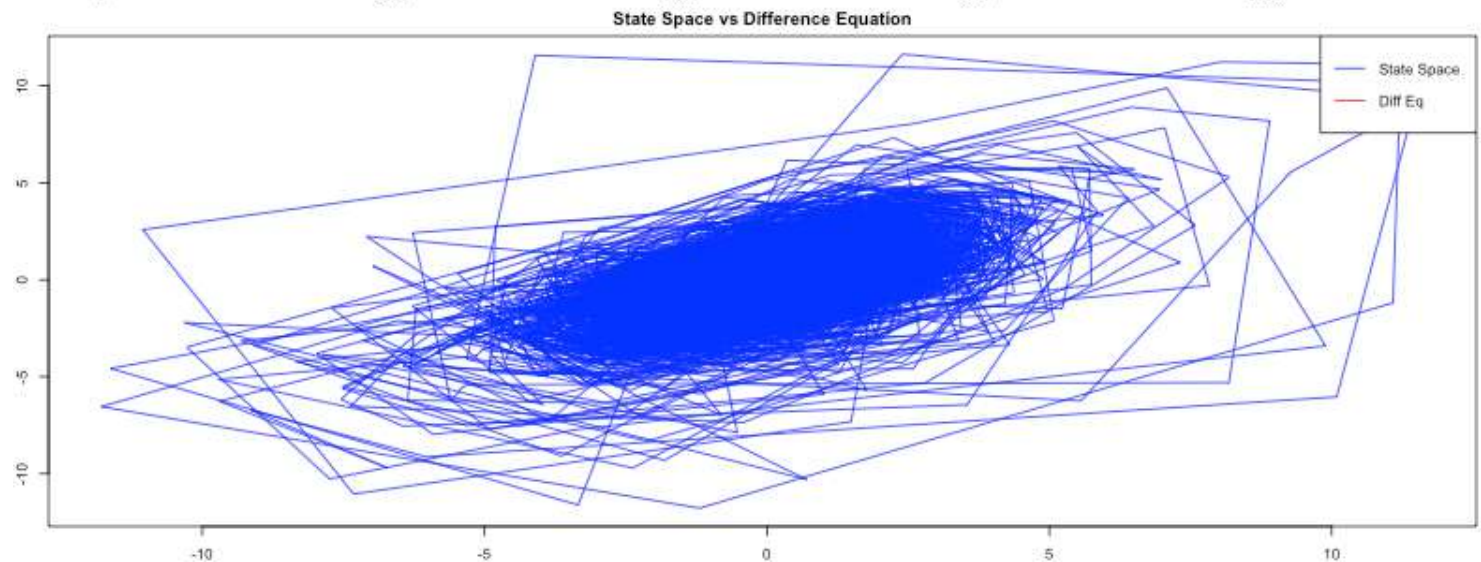
Comparison Results:
Max absolute difference: 16.12673
Mean squared error: 2.10382
> 
```

zsh
R Intera...
R: Insta...


S&P 500 Log Returns (%)
10
5
0
-5
-10
0
2000
4000
6000
8000


State Space Model Output
10
5
0
-5
-10
0
2000
4000
6000
8000


Difference Equation Output
10
5
0
-5
-10
0
2000
4000
6000
8000


State Space vs Difference Equation
State Space
Diff Eq
10
5
0
-5
-10
-10
-5
0
5
10